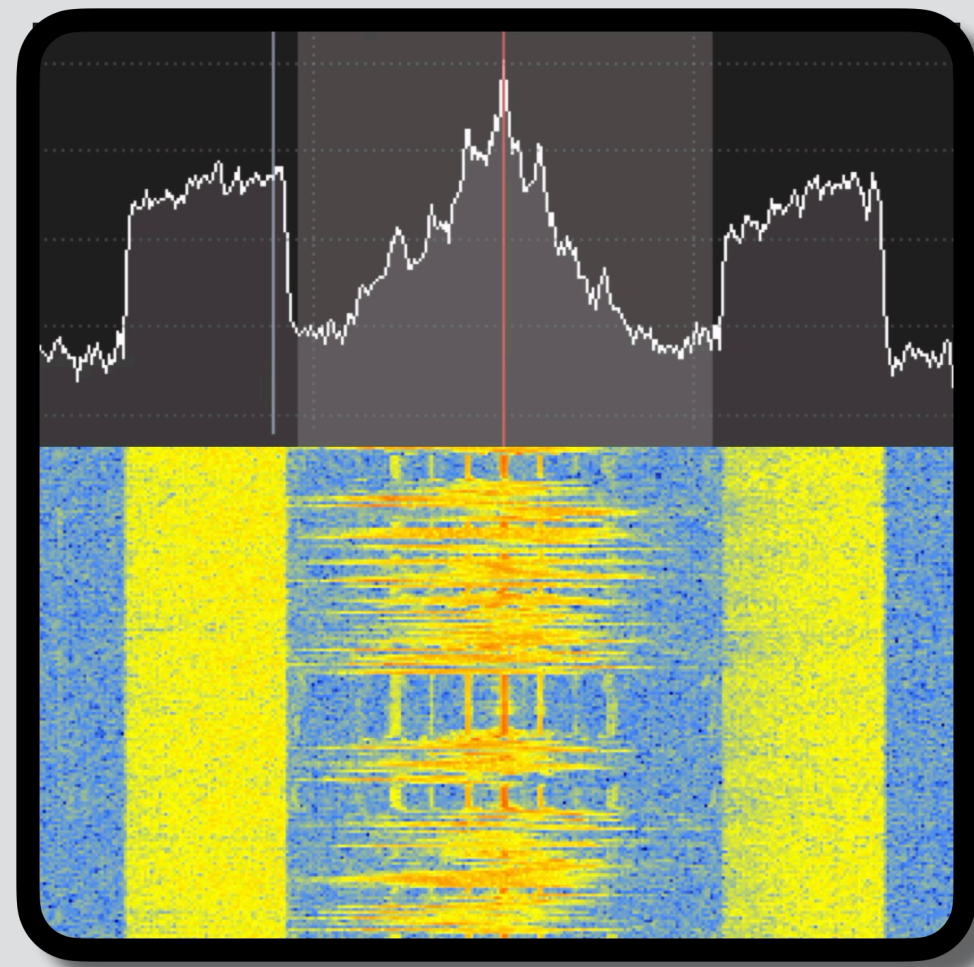
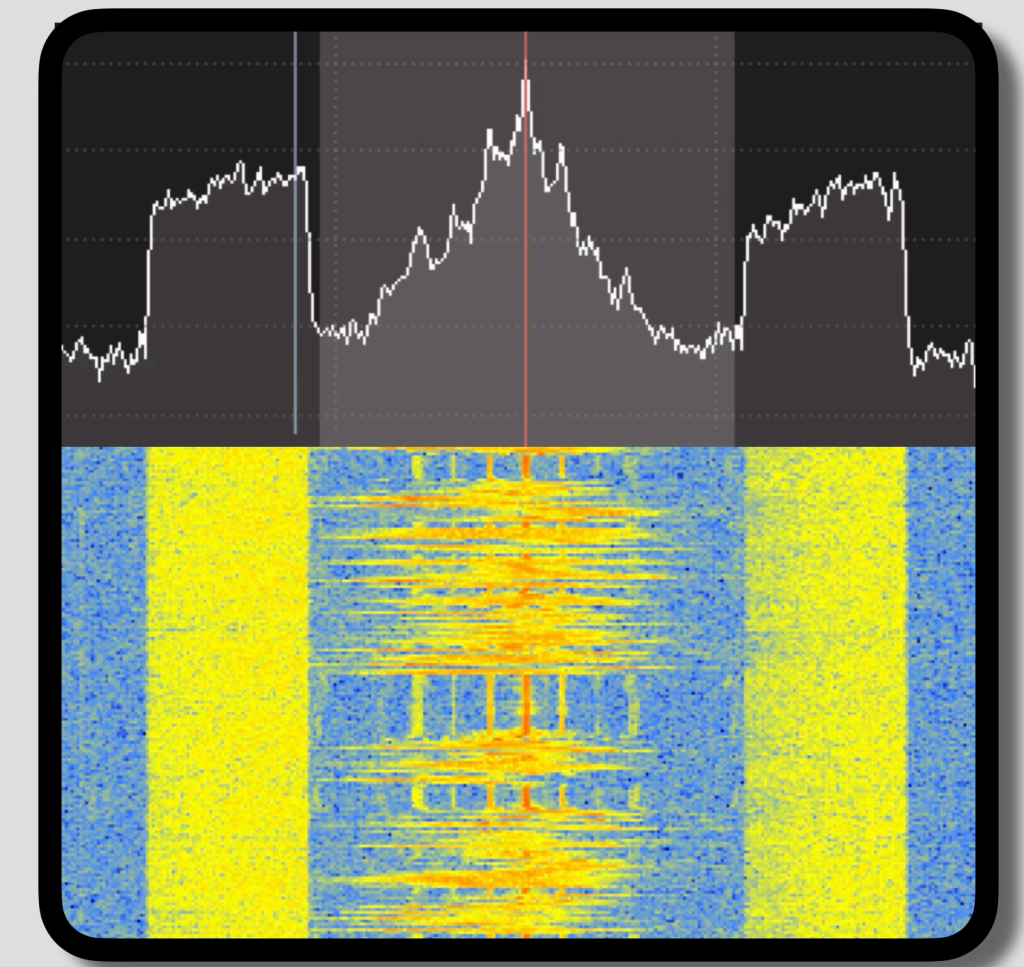




EE121

& Analog



Introduction to Digital Communication Systems

Lecture 9A

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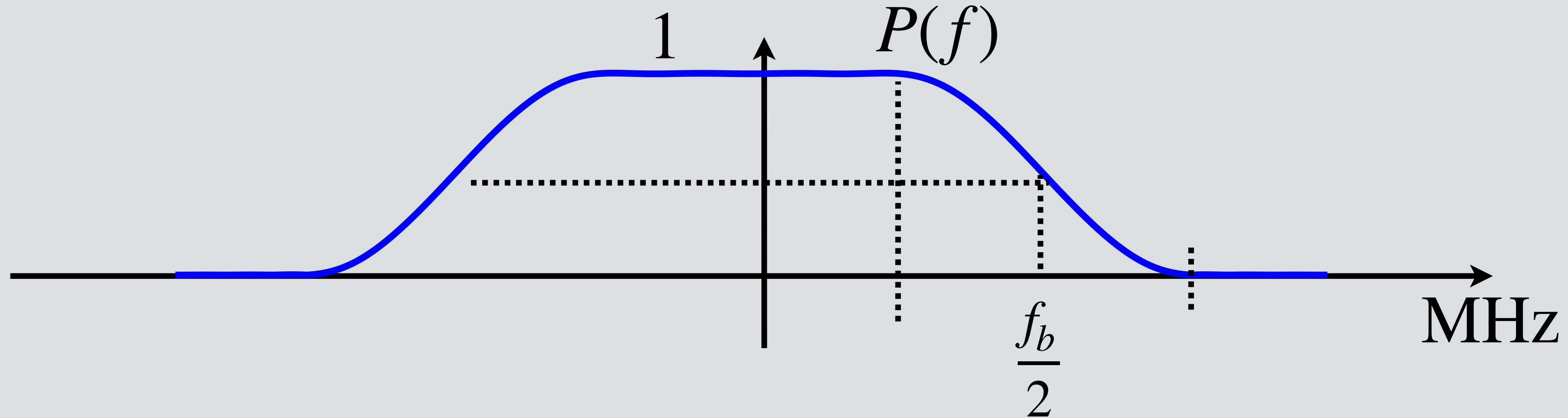
Today

- Last time:
 - Continue carrier Modulation
 - Lab 6
- Today:
 - Examples
 - Power Efficiency
 - Matched Filtering

Bandwidth Efficiency

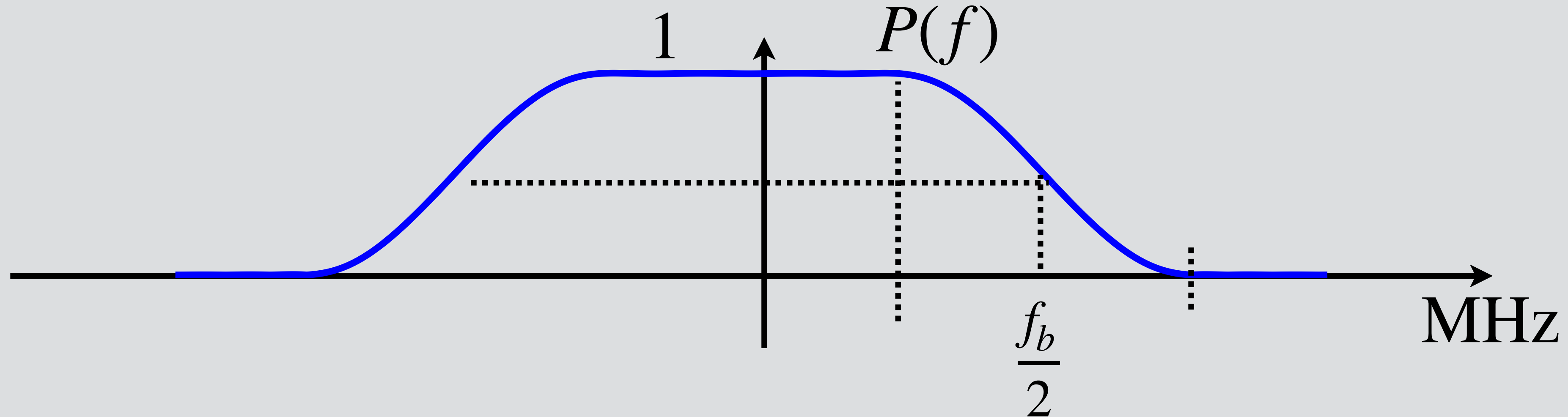
- Define bandwidth efficiency: $\eta_B = \log_2 M$ bits/symbols
- Minimum possible bandwidth is symbol / sec
- R_b = bits/seconds
- So, minimum BW, $B_{min} = \frac{R_b}{\eta_B}$, also baud, symbol-rate
- And actual bandwidth $B = (1 + \beta) \frac{R_b}{\eta_B}$

Example:



- Consider raised coding shaping above:
 - What is the BW using a RC pulse operating at 120Mbps using 64QAM and $\beta = 0.25$?

Example:



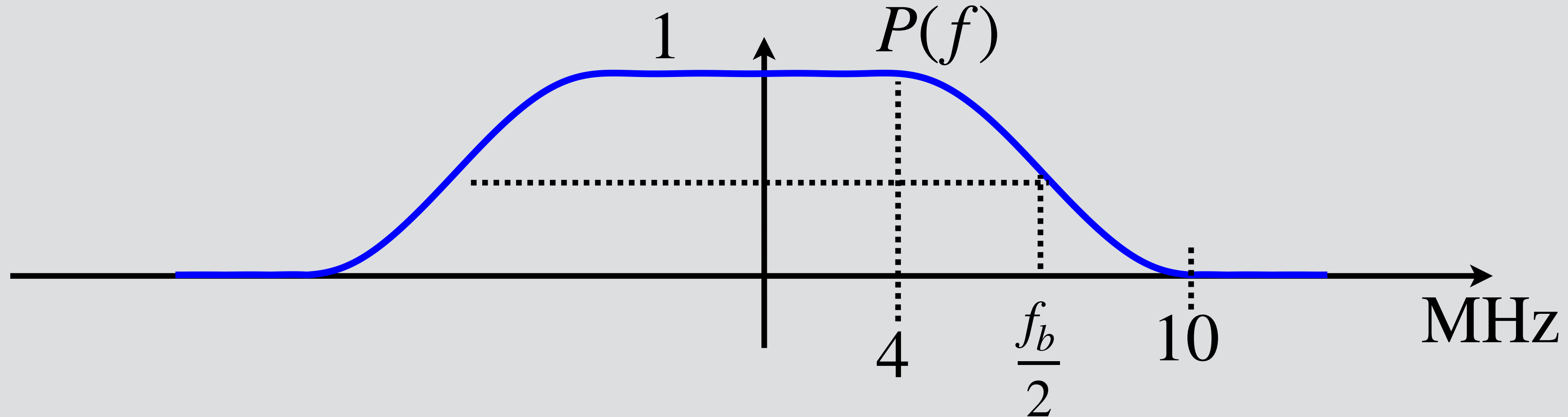
- Consider raised coding shaping above:
 - What is the BW using a RC pulse operating at 120Mbps using 64QAM and $\beta = 0.25$?

$$\eta_B = \log_2(M) = \log_2(64) = 6$$

$$B_{\min} = \frac{R_b}{\eta_B} = 120/6 = 20\text{MHz}$$

$$B = (1 + \beta)B_{\min} = 1.25 \times 20 = 25\text{MHz}$$

Example:

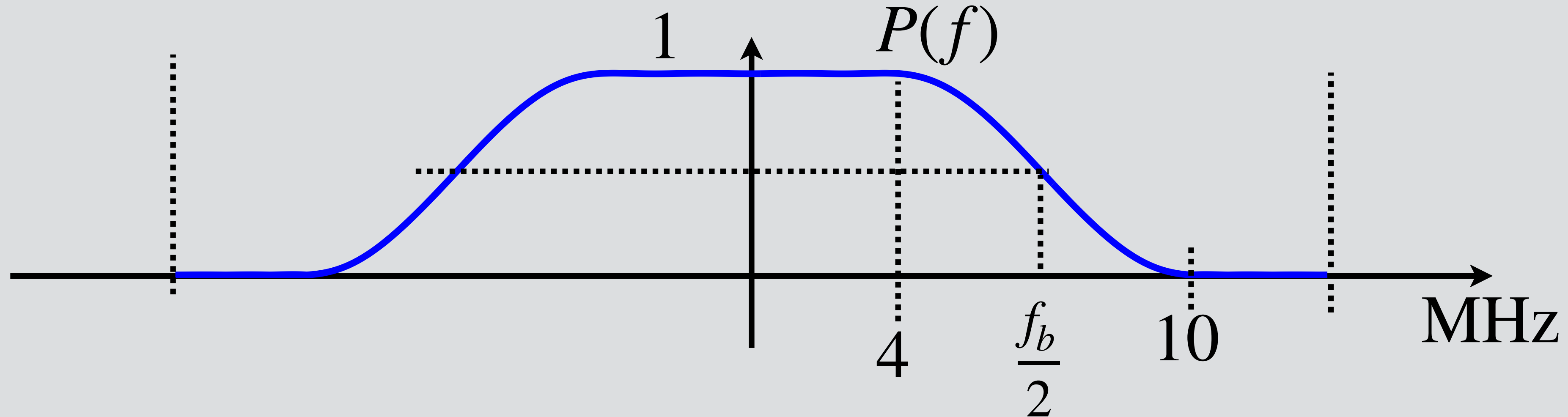


- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 56Mbps using 16QAM?

$$p(t) = \text{sinc} \left(\frac{t}{T_b} \right) \frac{\cos(\pi \beta t / T_b)}{1 - \left(\frac{2\beta t}{T} \right)^2}$$

Zero-crossing every: $\Rightarrow kT_b \Rightarrow f_b/k$
 $\Rightarrow k \frac{T_b}{\beta} \Rightarrow \beta f_b/k$

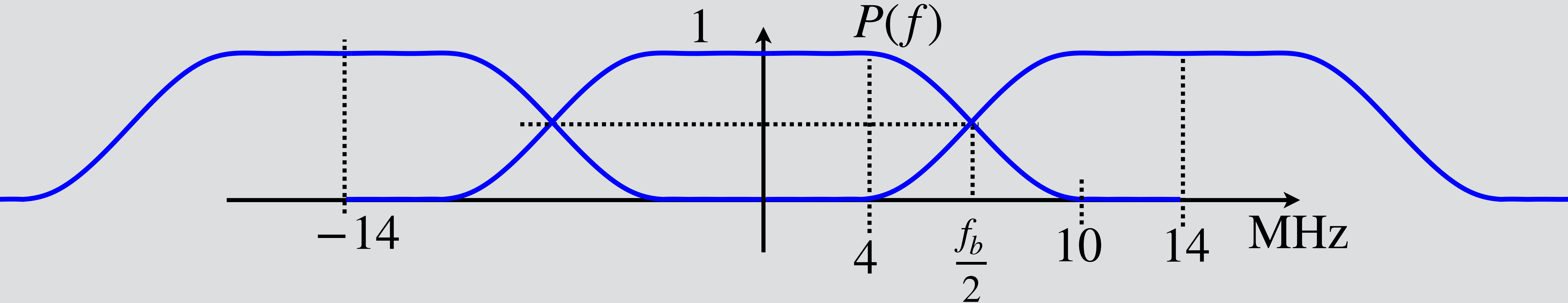
Example:



- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 56Mbps using 16QAM?

Symbol-rate: $\tilde{f}_b = \frac{56}{4} = 14$ Mbaud RC: $f_b = 14$

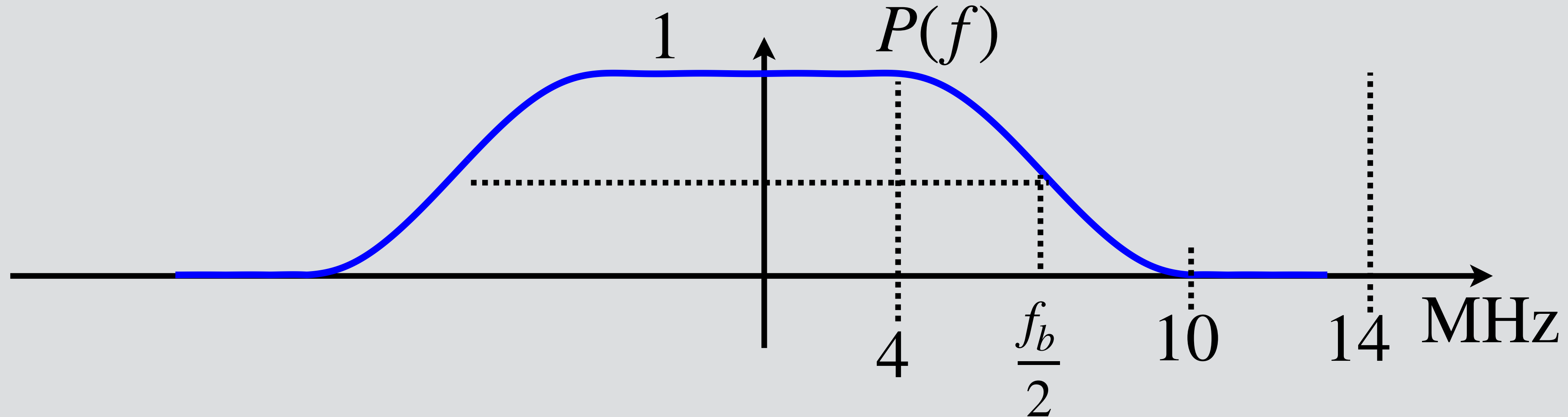
Example:



- Consider raised coding shaping above. True/False:
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Example:



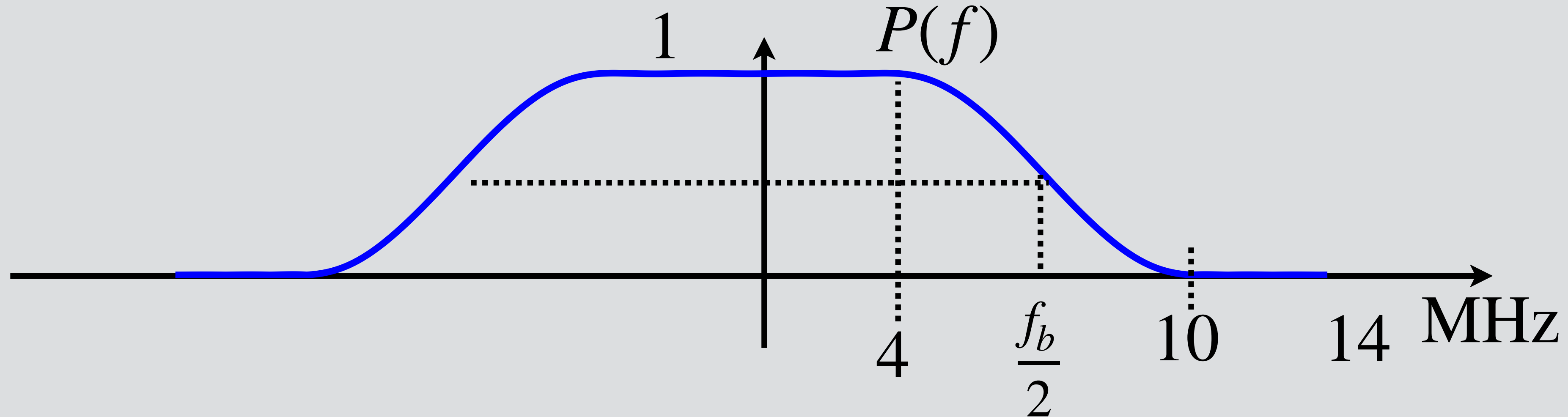
- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 56Mbps using 16QAM?

Symbol-rate: $\tilde{f}_b = \frac{56}{4} = 14$ Mbaud RC: $f_b = 14$

$$\Rightarrow \frac{f_b}{k} = \frac{14}{k} \quad k = 1 \Rightarrow 14$$

$$\beta = \frac{3}{7} \Rightarrow \frac{3 \cdot 14}{7k} \quad k = 1 \Rightarrow \frac{3}{7} 14 = 6$$

Example:



- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 21Mbps using 8PSK?

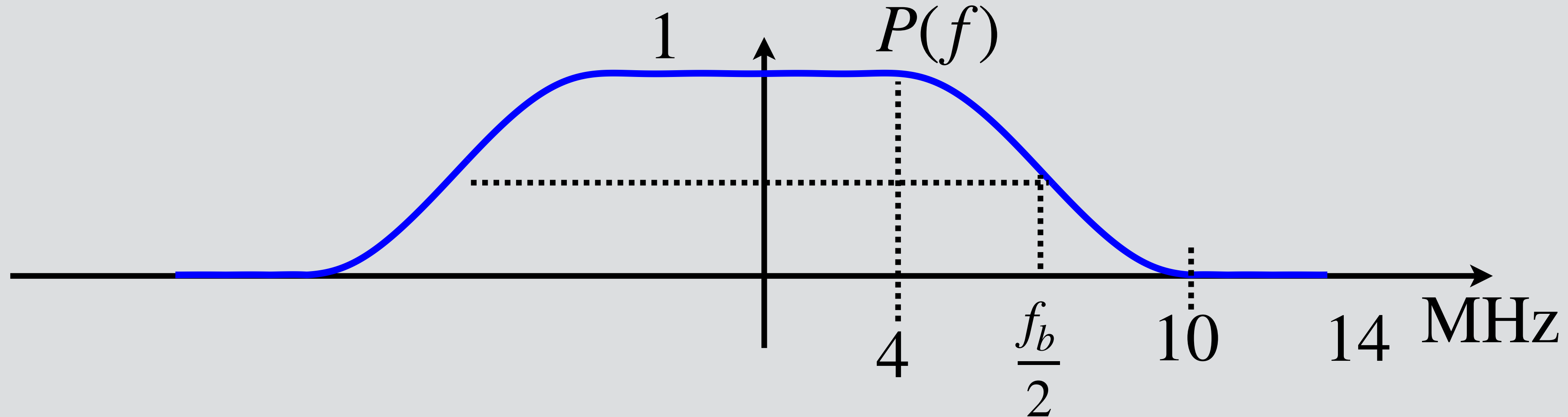
Symbol-rate:

$$\text{RC: } f_b = 14$$

$$\Rightarrow \frac{14}{k}$$

$$\Rightarrow \frac{3 \cdot 14}{7k}$$

Example:



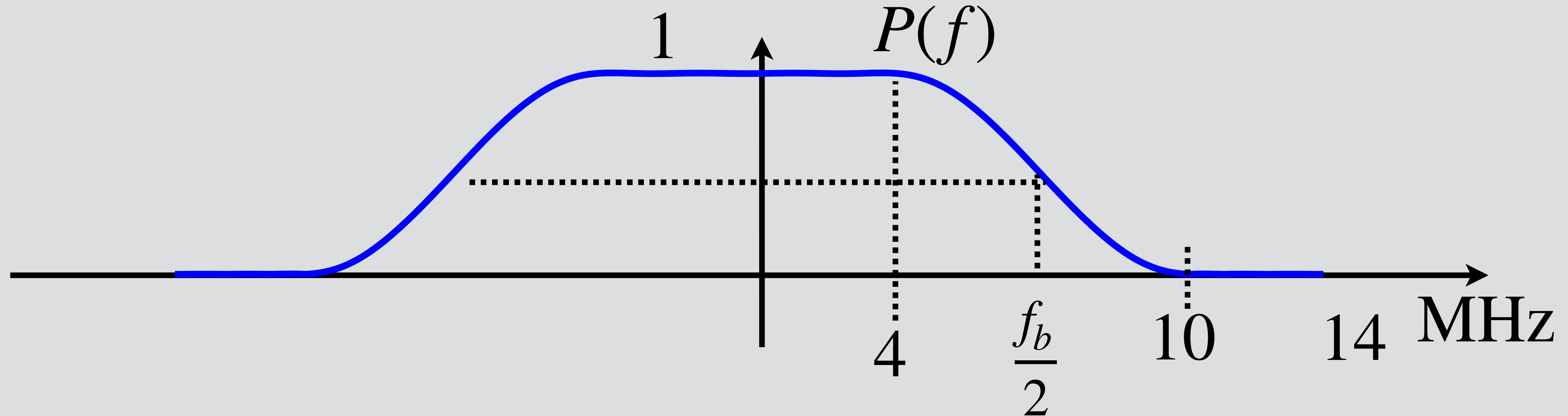
- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 21Mbps using 8PSK?

Symbol-rate: $\tilde{f}_b = \frac{21}{3} = 7$ Mbaud RC: $f_b = 14$

$$\Rightarrow \frac{14}{k} \quad k = 2 \Rightarrow 7$$

$$\Rightarrow \frac{3 \cdot 14}{7k}$$

Example:



- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 18Mbps using 8PSK?

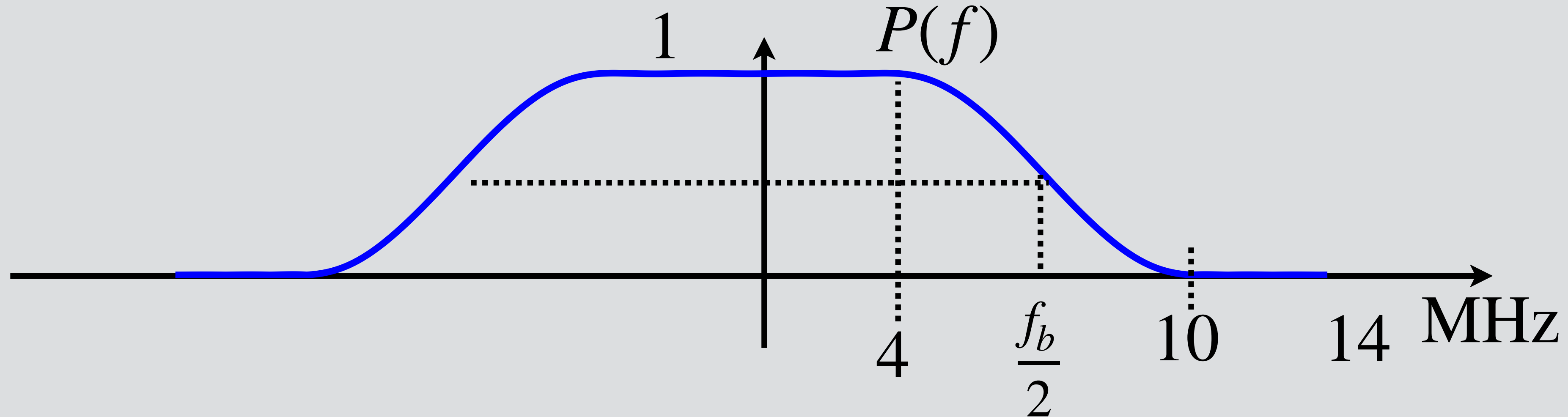
Symbol-rate:

$$\text{RC: } f_b = 14$$

$$\Rightarrow \frac{14}{k}$$

$$\Rightarrow \frac{3 \cdot 14}{7k}$$

Example:



- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 18Mbps using 8PSK?

Symbol-rate: $\tilde{f}_b = \frac{18}{3} = 6$ Mbaud

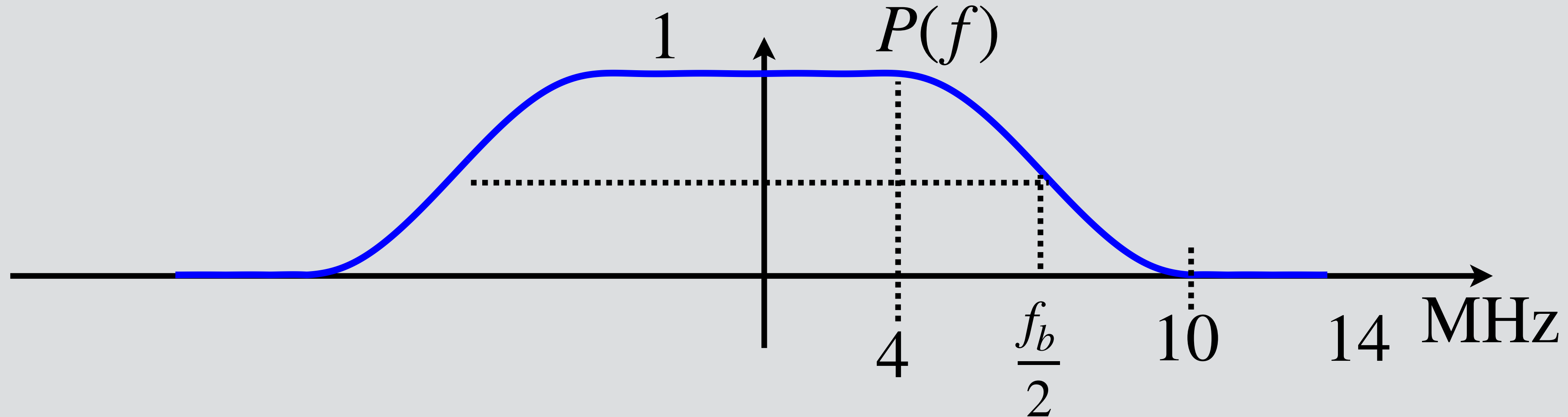
RC: $f_b = 14$

$$\Rightarrow \frac{14}{k}$$

$$\Rightarrow \frac{3 \cdot 14}{7k}$$

$$k = 1 \Rightarrow \frac{3}{7} 14 = 6$$

Example:



- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 25Mbps using QPSK?

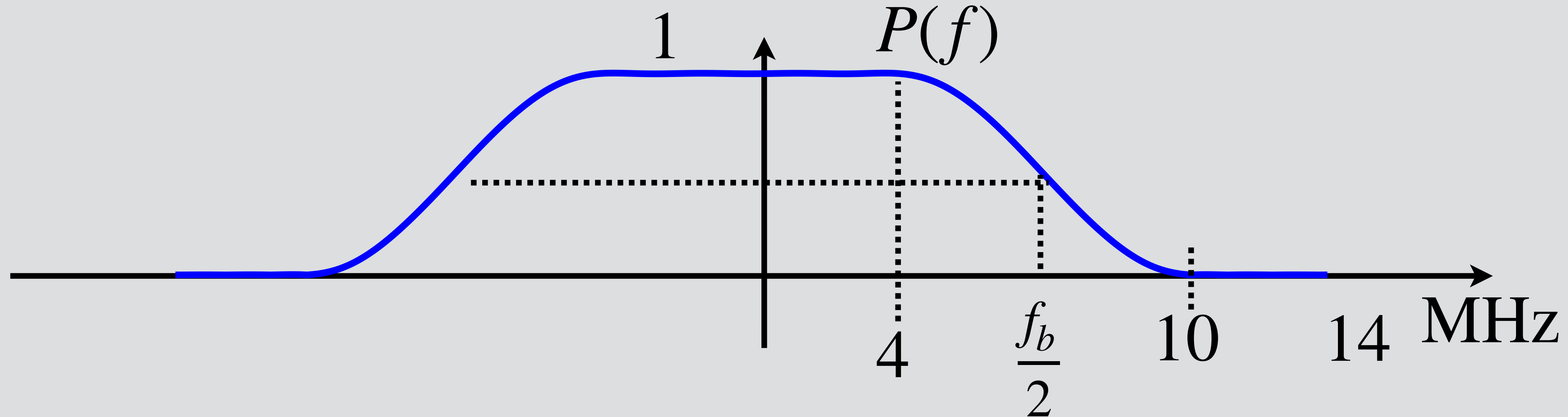
Symbol-rate:

$$\text{RC: } f_b = 14$$

$$\Rightarrow \frac{14}{k}$$

$$\Rightarrow \frac{3 \cdot 14}{7k}$$

Example:



- Consider raised coding shaping above. True/False:
 - Pulse has zero ISI at bitrate of 25Mbps using QPSK?

Symbol-rate: $\frac{1}{T} = \frac{25}{2} = 12.5$ Mbaud RC: $f_b = 14$

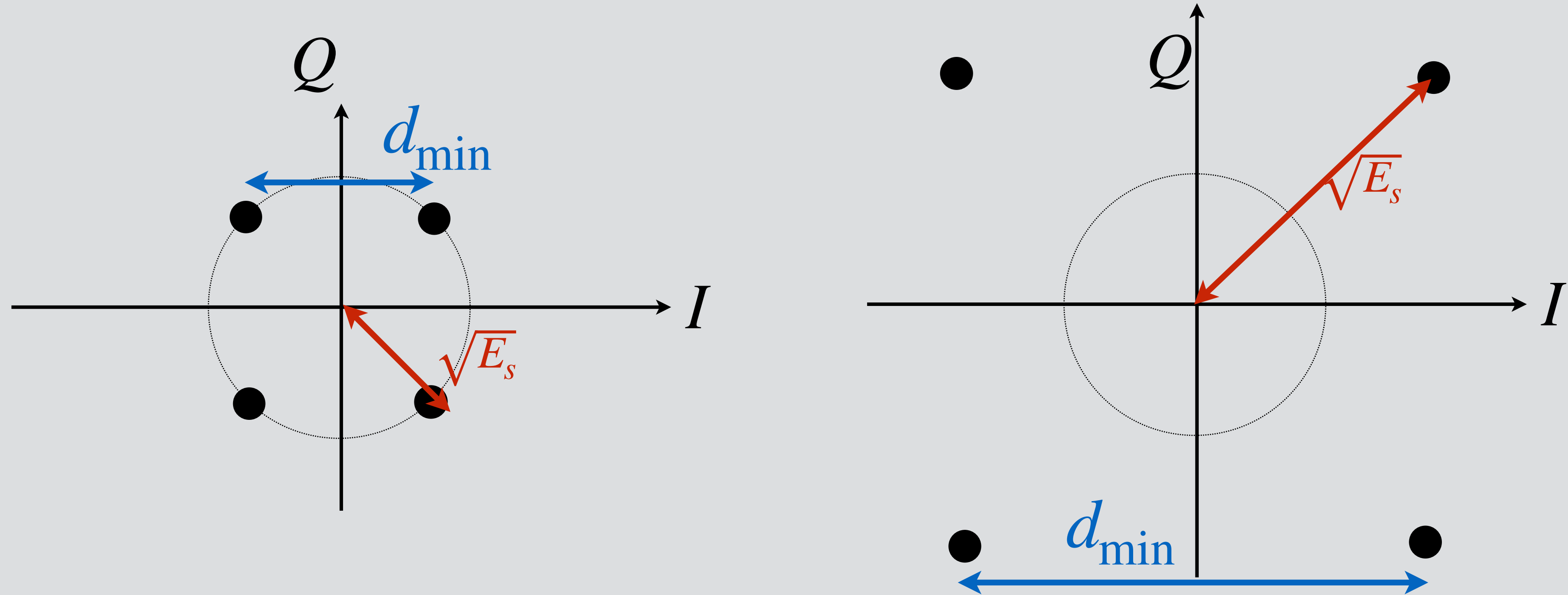
$\Rightarrow \frac{14}{K}$

$\Rightarrow \frac{3 \cdot 14}{7K}$

Power-Bandwidth

- Bandwidth efficiency increases with M (-array)
 - BPSK = 1bits/sym, QPSK = 2bits/sym, 16QAM = 4 bit/sym
- Why not keep increasing size?
 - Noise
 - Power

Minimum Distance and Energy / symbol



- $d_{\min}$ is minimum distance between points
 - Measures sensitivity to noise
- E_s is mean energy per symbol (mean squared distance)

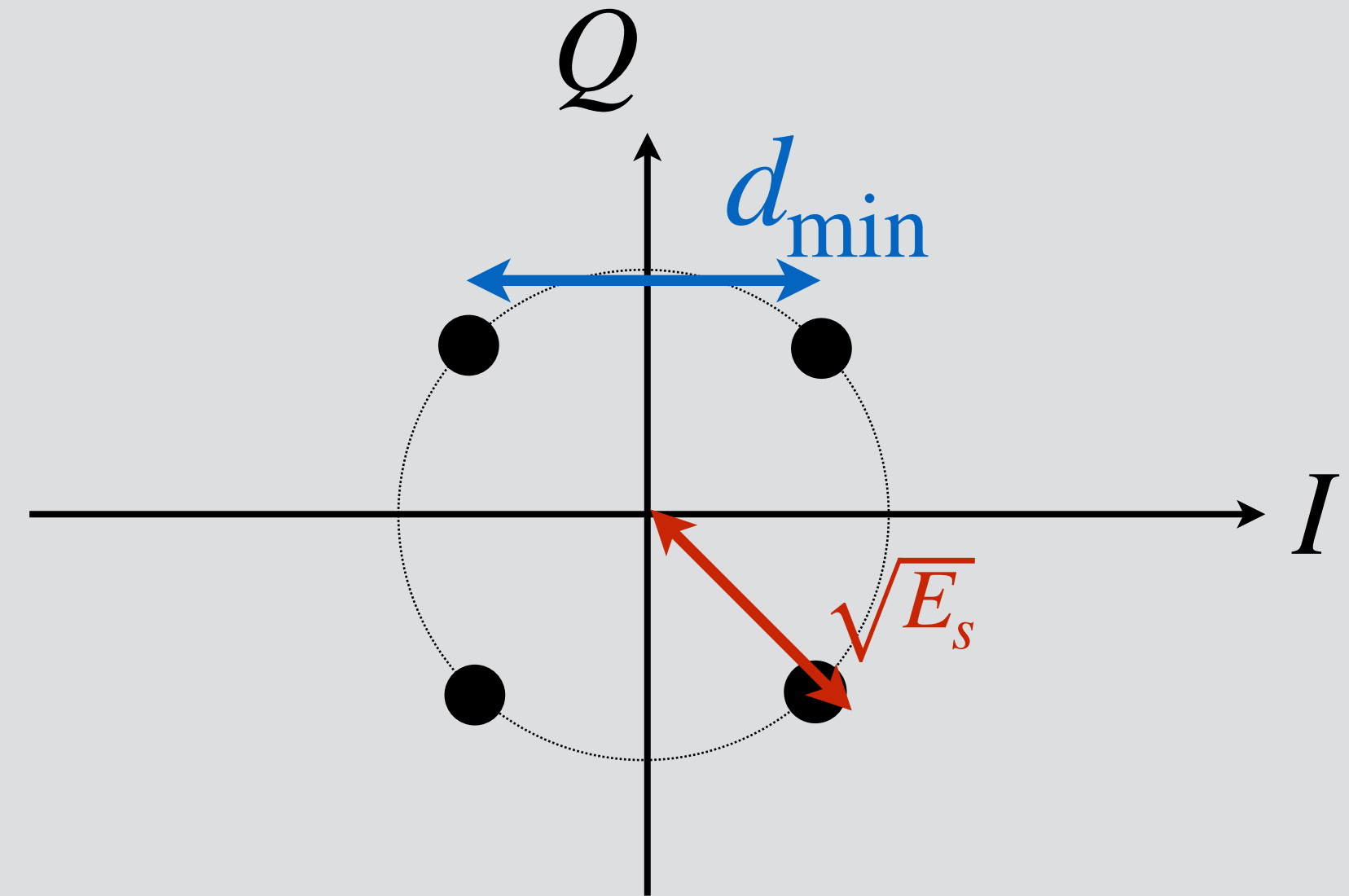
Minimum Distance and Energy / symbol

$$p_k(t) = (a_k + jb_k)e^{j2\pi f_c t/T_b}$$

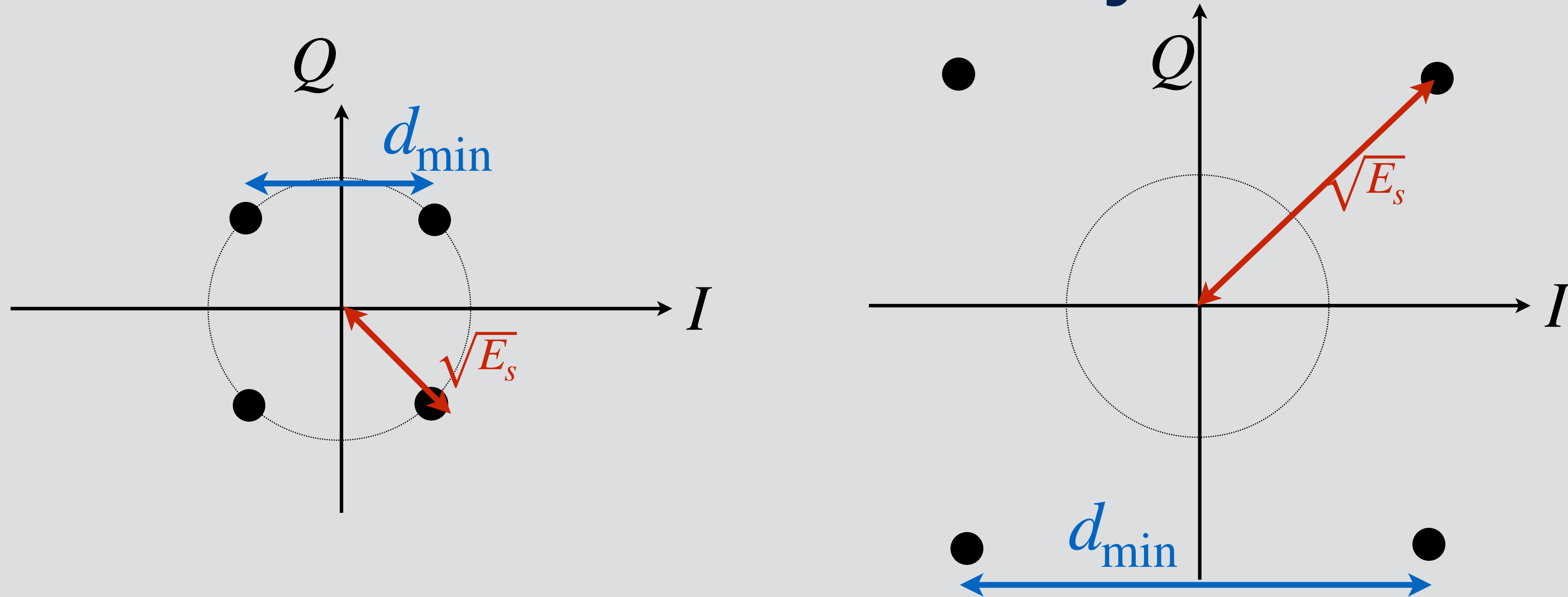
$$\begin{aligned} d_{\min} &= \min_{k \neq n} ||p_k(t) - p_n(t)||_2 \\ &= \min_{k \neq n} \left((a_k - a_n)^2 + (b_k - b_n)^2 \right)^{1/2} \end{aligned}$$

$$E_s = \frac{1}{M} \sum_k ||p_k(t)||_2^2 = \frac{1}{M} \sum_k (a_k^2 + b_k^2)$$

$$E_b = \frac{E_s}{\log_2 M} \quad \text{Mean energy per bit}$$



Power Efficiency

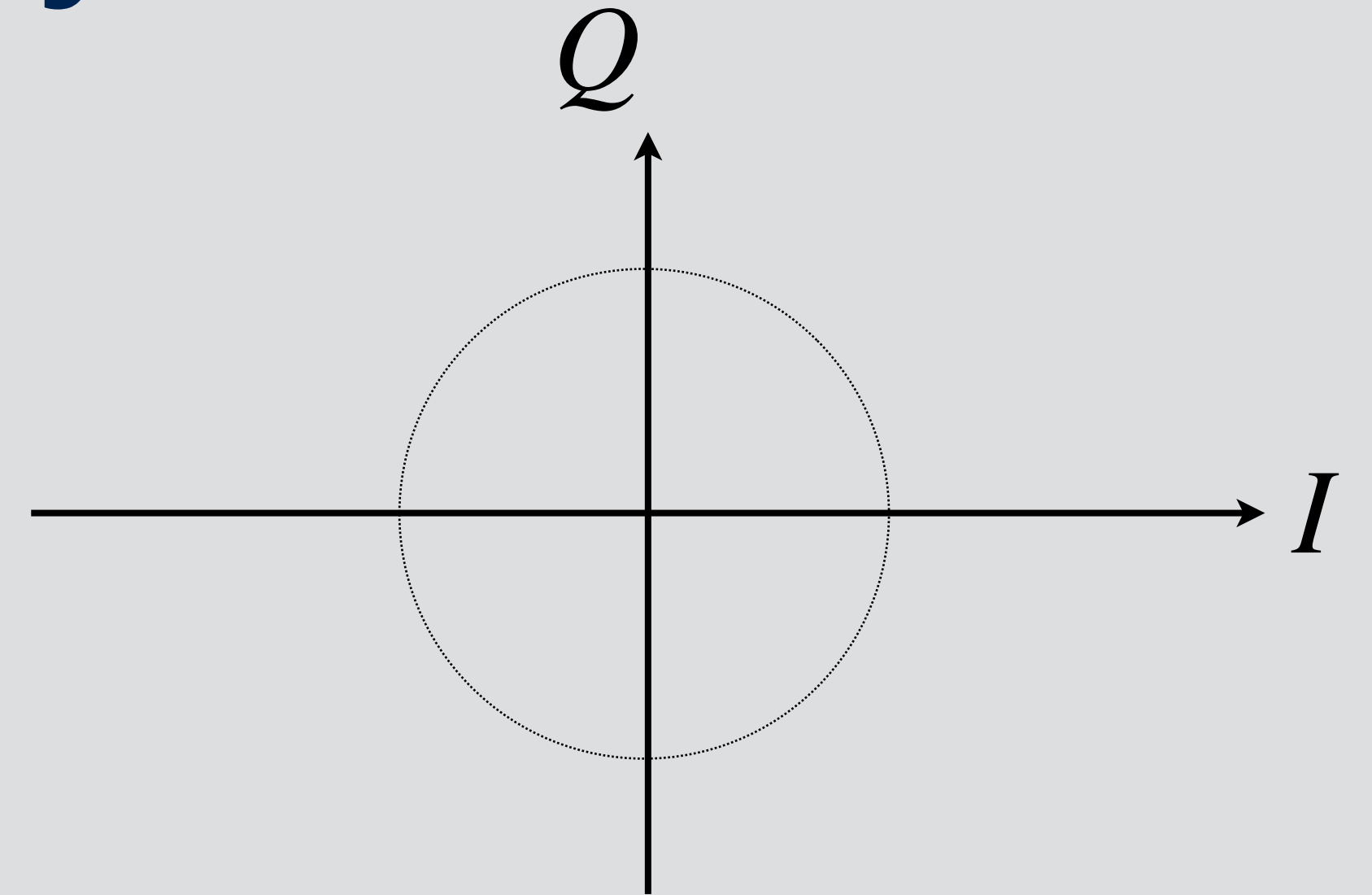


- $d_{\min}$ doubles
- E_s/E_b quadruple

$$\Rightarrow \eta_P = \frac{d_{\min}^2}{E_b} \quad \text{Is power efficiency is scaling invariant}$$

Power efficiency

OOK =



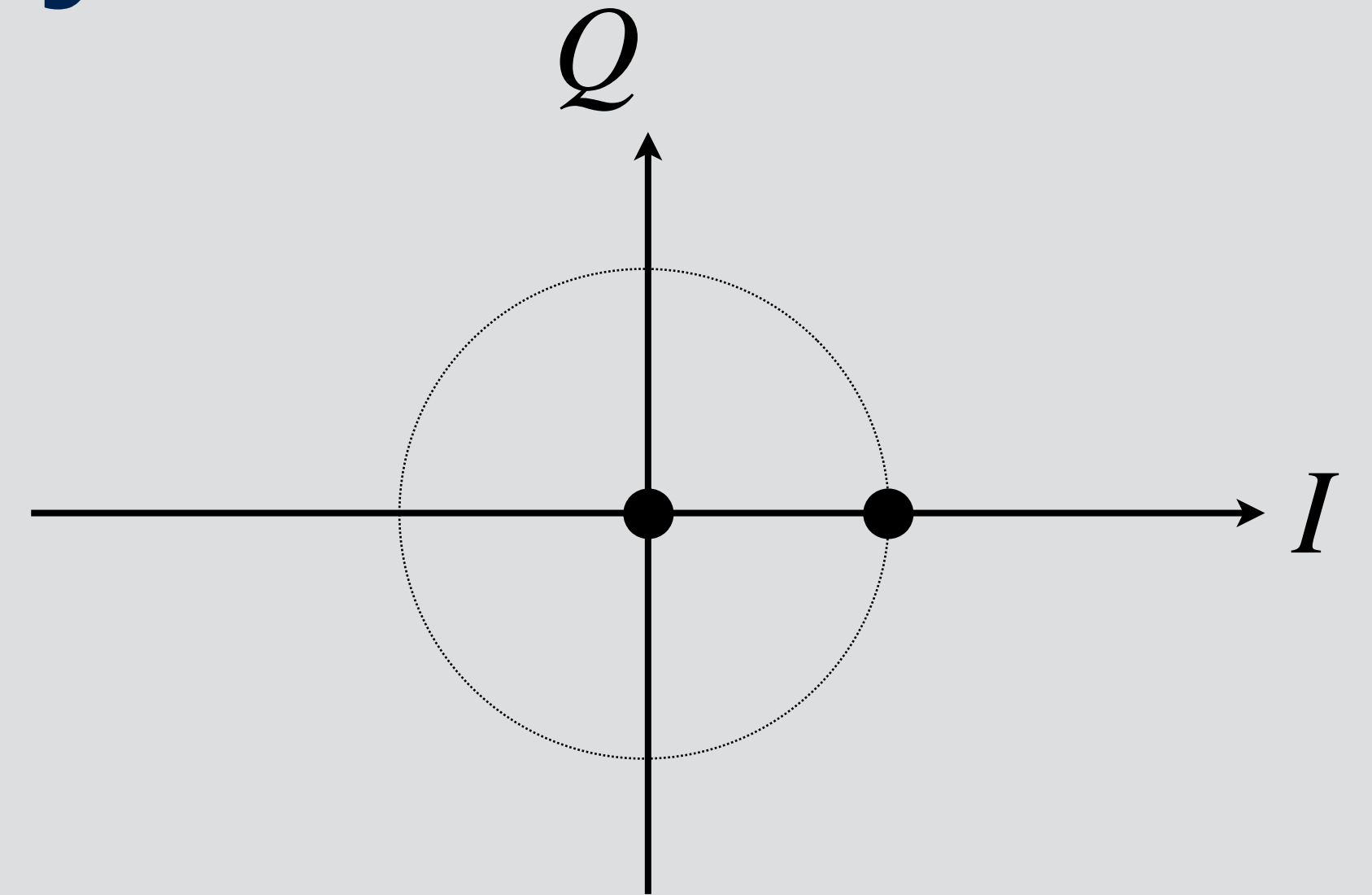
Power efficiency

$$\text{OOK} = \frac{d_{\min}^2}{E_b} = \frac{A^2}{A^2/2} = 2$$

$$d_{\min} = A$$

$$E_s = \frac{1}{M} \sum_k (a_k^2 + b_k^2) = \frac{1}{2}0 + \frac{1}{2}A^2 = \frac{A^2}{2}$$

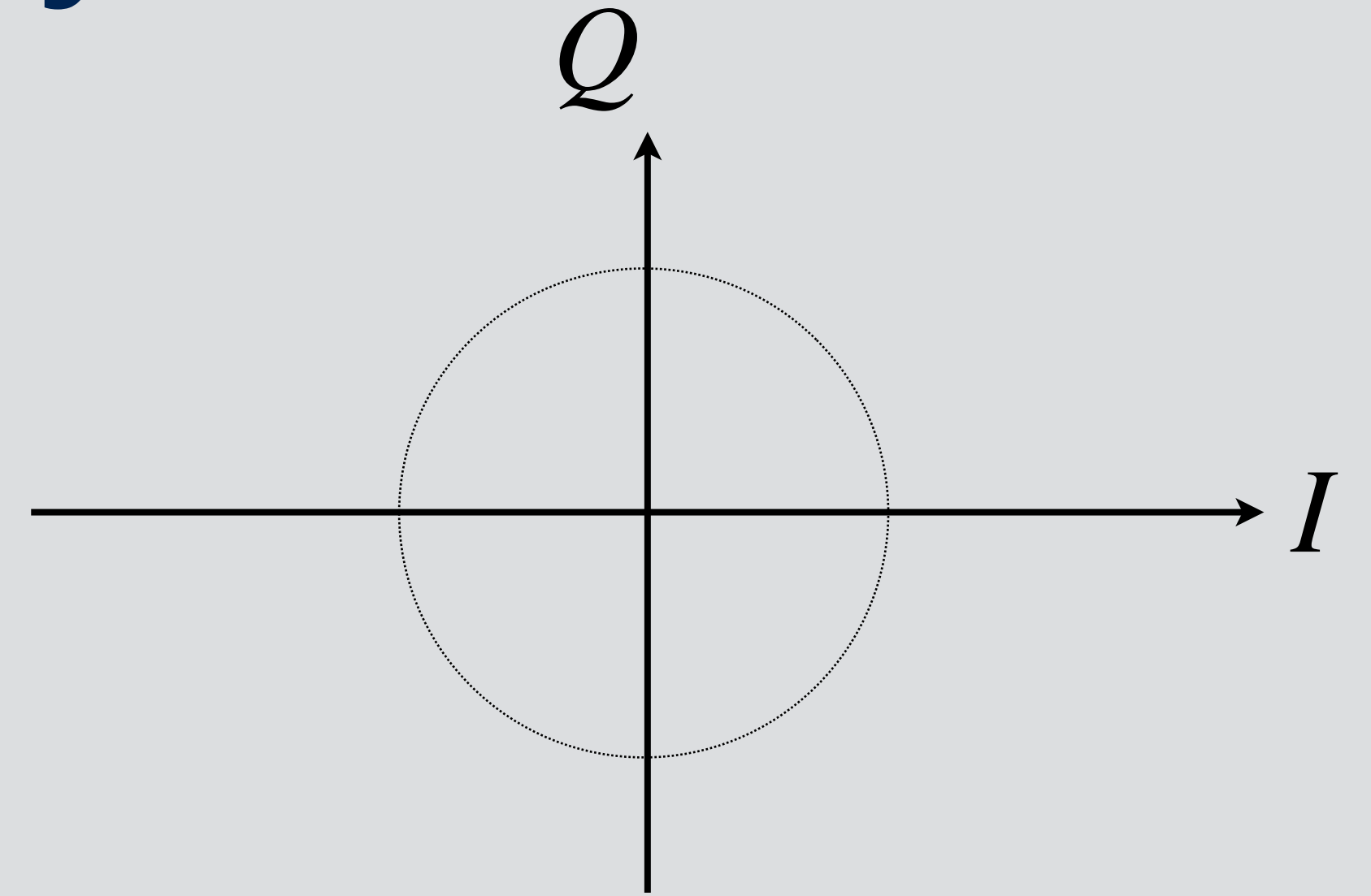
$$E_b = E_s / \log_2(M) = \frac{A^2}{2} / 1 = \frac{A^2}{2}$$



Power efficiency

OOK = 2

BPSK =



Power efficiency

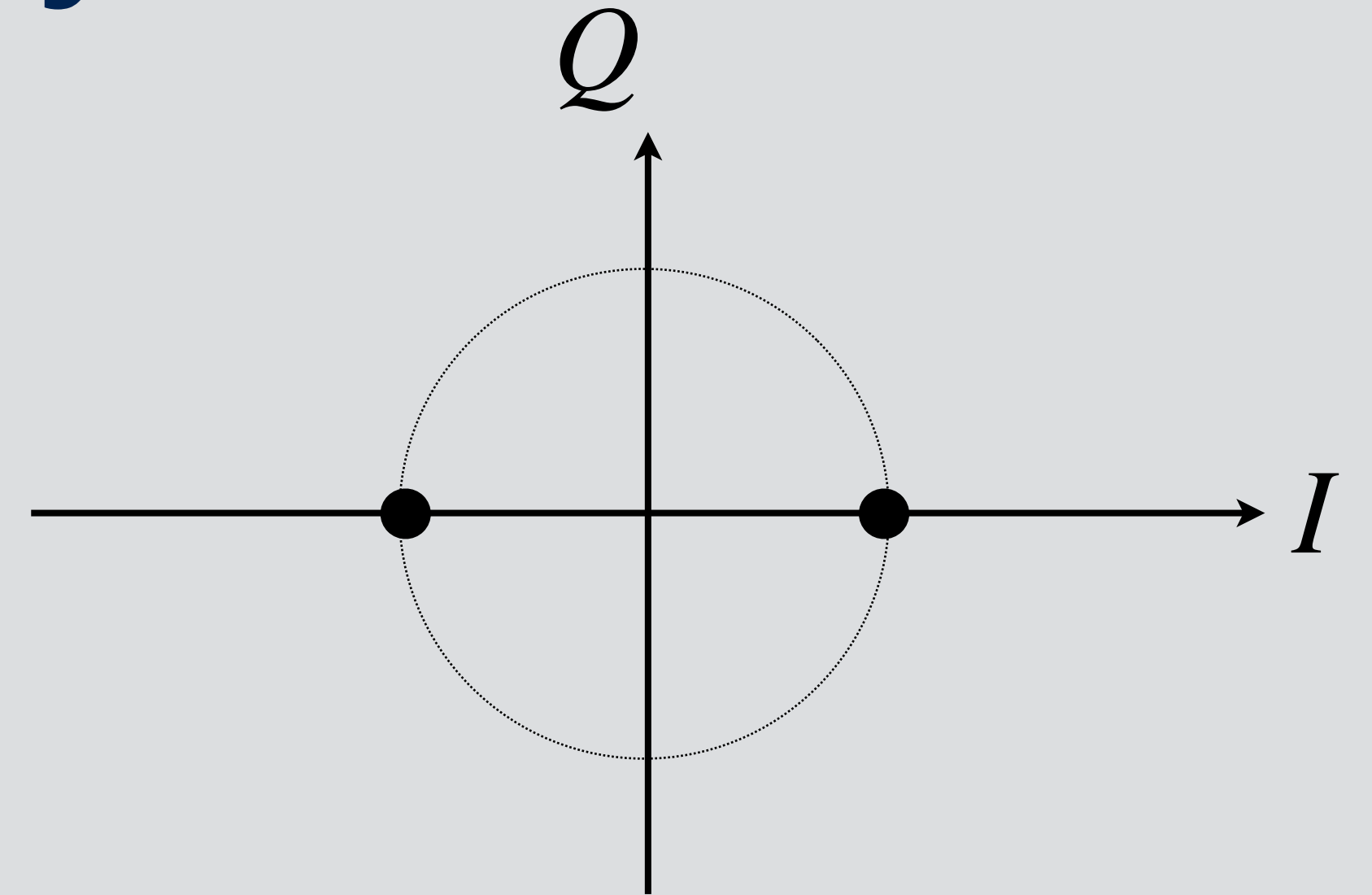
$$\text{OOK} = 2$$

$$\text{BPSK} = \frac{d_{\min}^2}{E_b} = 2^2/1 = 4$$

$$d_{\min} = 2$$

$$E_s = 1$$

$$E_b = 1$$

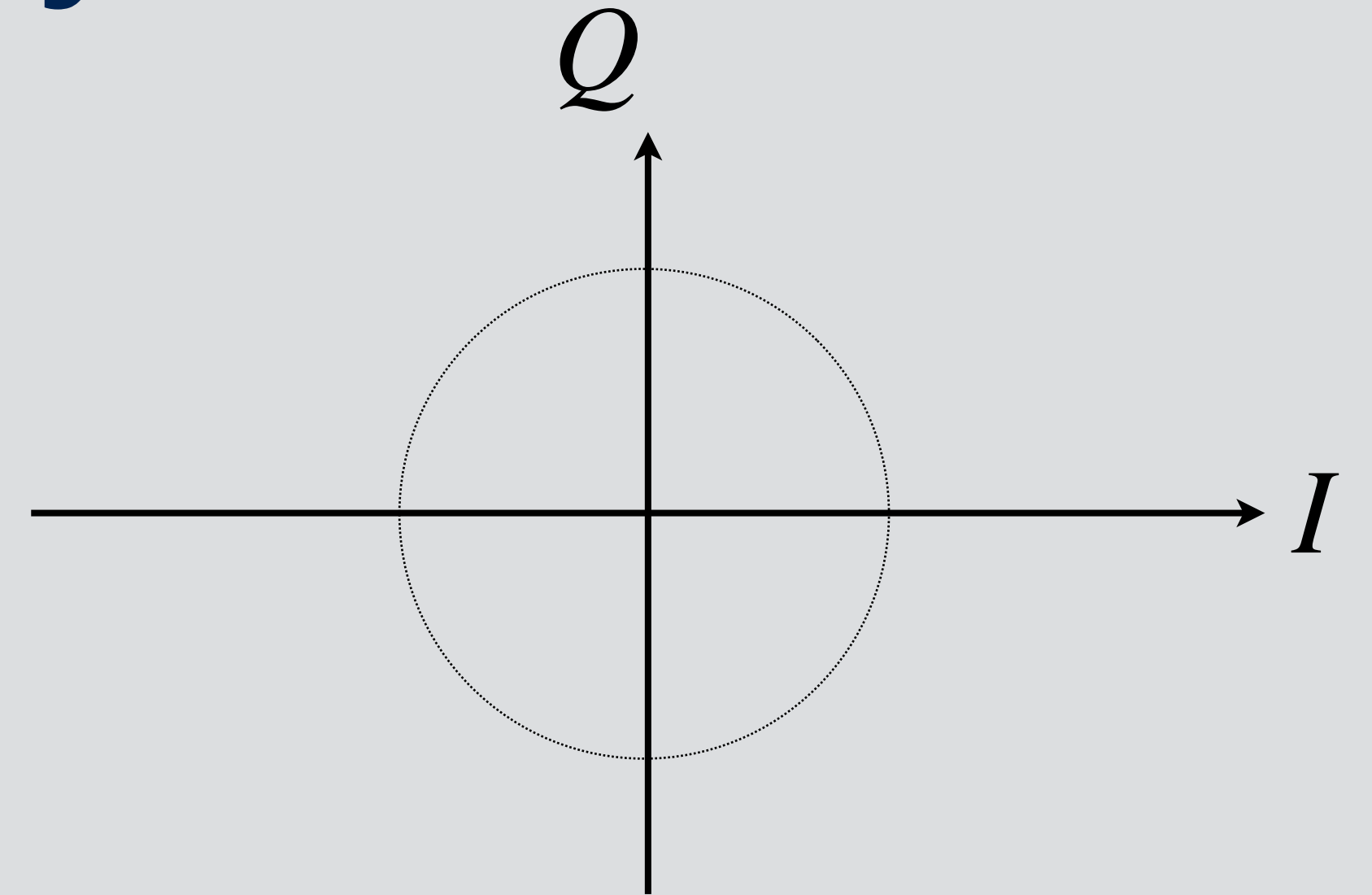


Power efficiency

OOK = 2

BPSK = 4

4-PAM =



Power efficiency

$$\text{OOK} = 2$$

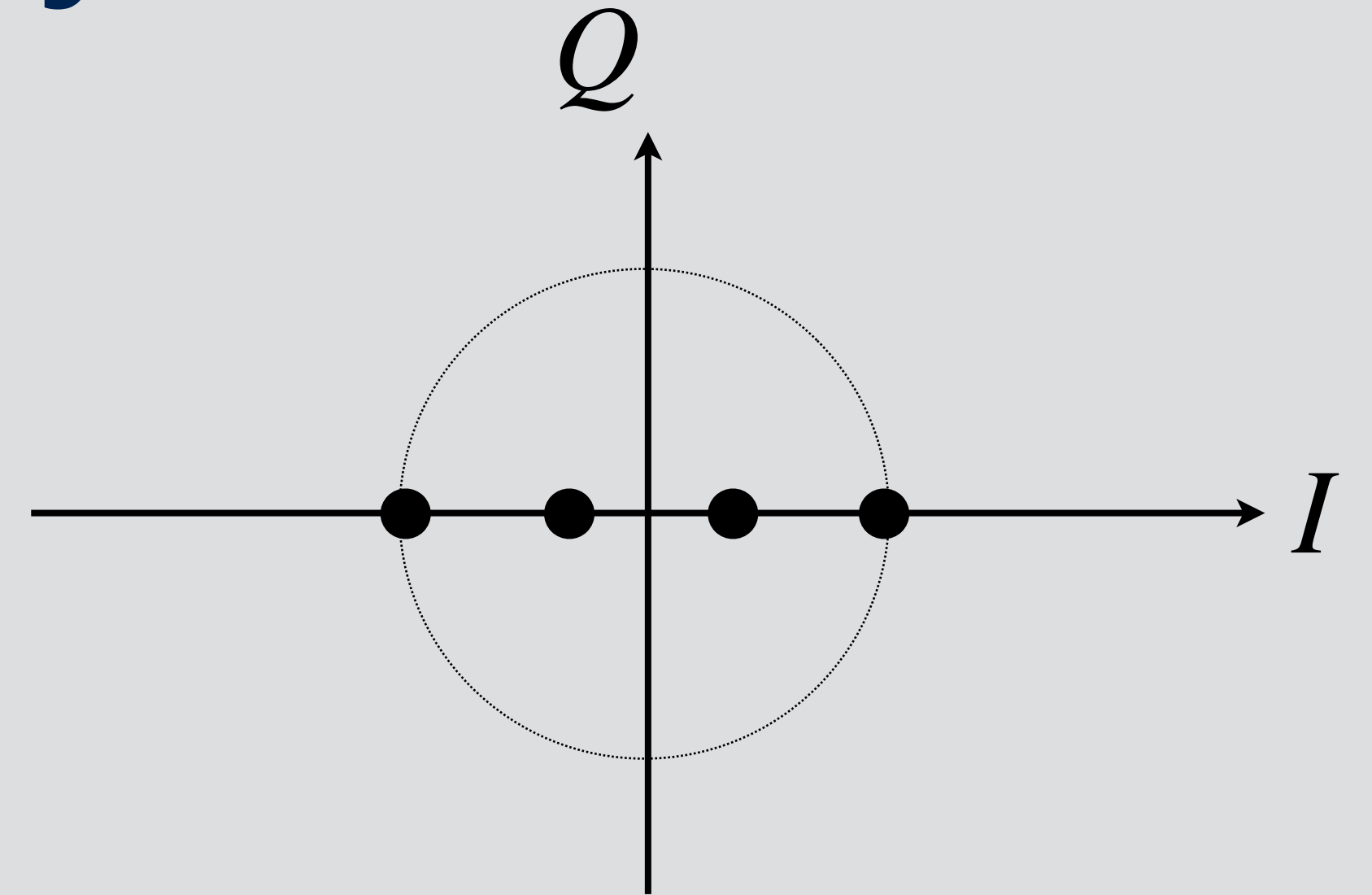
$$\text{BPSK} = 4$$

$$\text{4-PAM} = \frac{d_{\min}^2}{E_b} = \frac{4}{9} \frac{18}{5} = \frac{8}{5} = 1.6$$

$$d_{\min} = 2/3$$

$$E_s = \frac{1}{4} \left(2 \cdot 1 + 2 \cdot \frac{1}{3^2} \right) = \frac{5}{9}$$

$$E_b = E_s/2 = 5/18$$



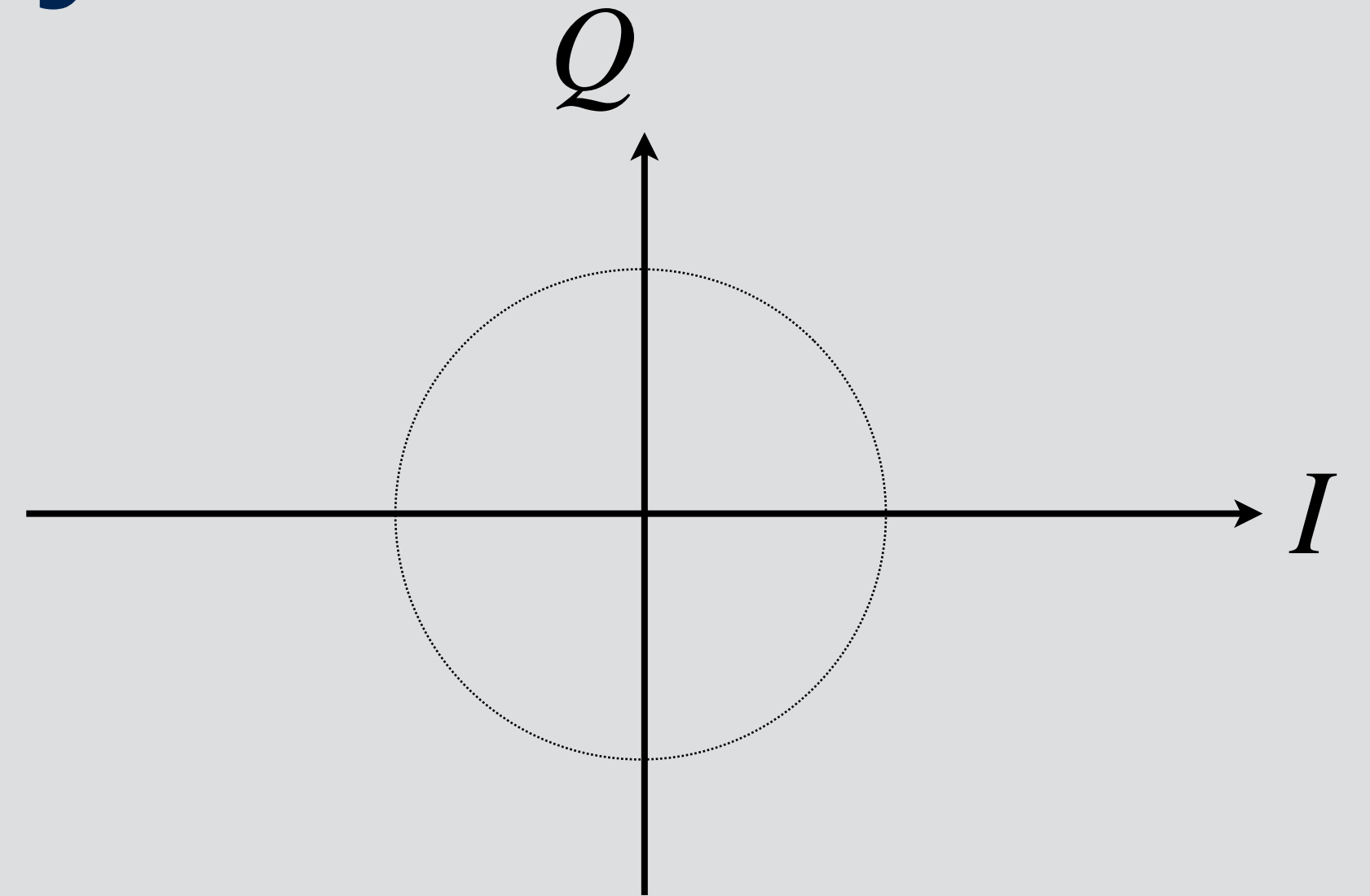
Power efficiency

OOK = 2

BPSK = 4

M-PAM = 1.6

QPSK/4QAM =



Power efficiency

$$\text{OOK} = 2$$

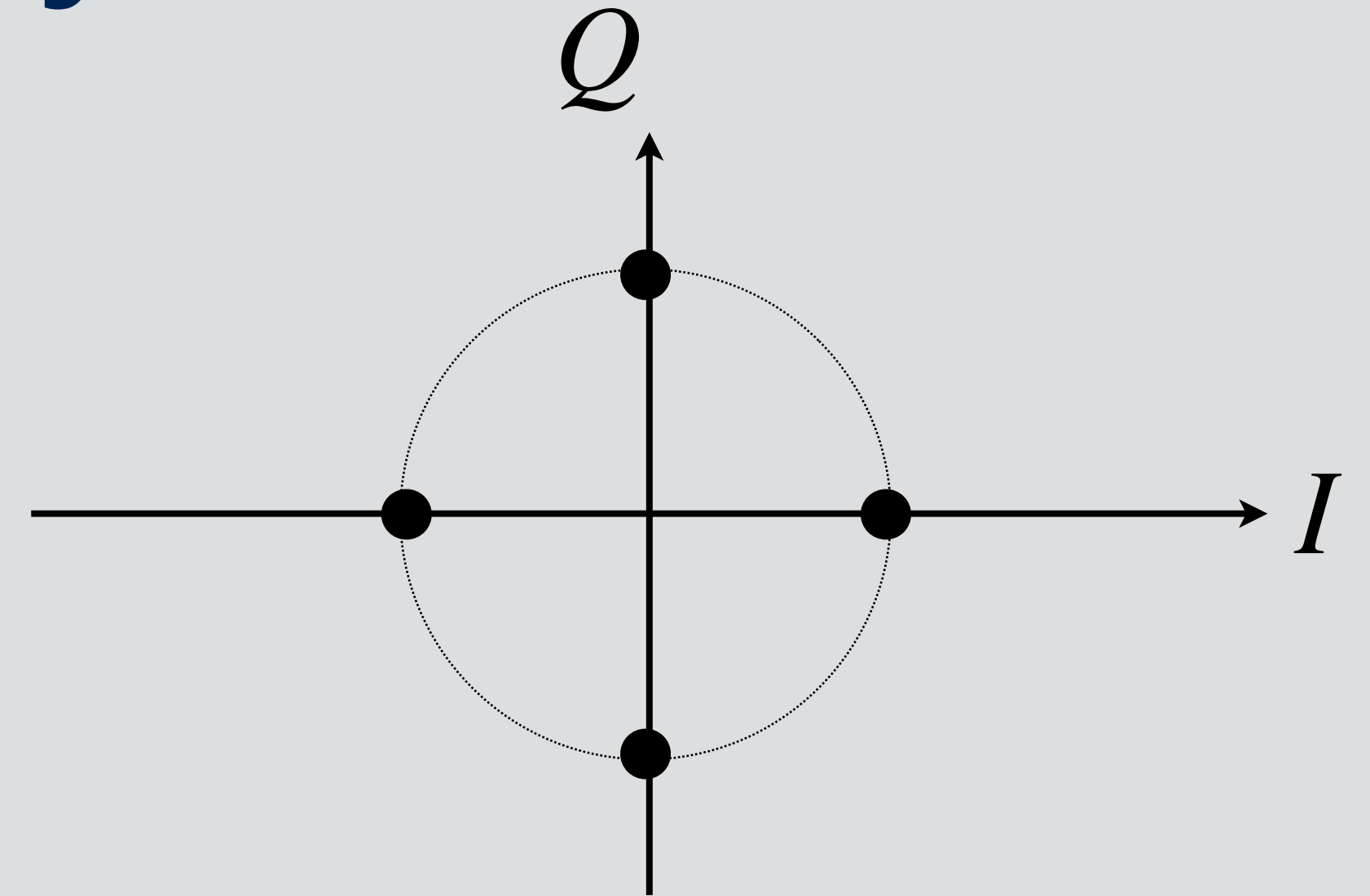
$$\text{BPSK} = 2$$

$$\text{M-PAM} = 1.6$$

$$\text{QPSK/4QAM} = \frac{d_{\min}^2}{E_b} = \frac{2}{1/2} = 4$$

$$d_{\min} = \sqrt{2}$$

$$E_s = \frac{1}{4}(1 + 1 + 1 + 1) = 1 \quad E_b = E_s / \log_2(4) = 1/2$$



Link Parameters and Power Efficiency

R_b — bit rate

P_{RX} — received power

N_0 — noise PSD

$$\Rightarrow E_b = \frac{P_{RX}}{R_b} \quad \frac{\text{W}}{\text{bits/s}} = \frac{\text{J/s}}{\text{bits/s}} = \frac{\text{J}}{\text{bits}}$$

$$\Rightarrow \frac{E_b}{N_0} \quad \text{Measure of SNR}$$

$$\Rightarrow \eta_P \frac{E_b}{N_0} = \frac{d_{\min}^2}{N_0} \quad \text{Measure of reliability}$$

For a given reliability, more power is needed for less power efficiency

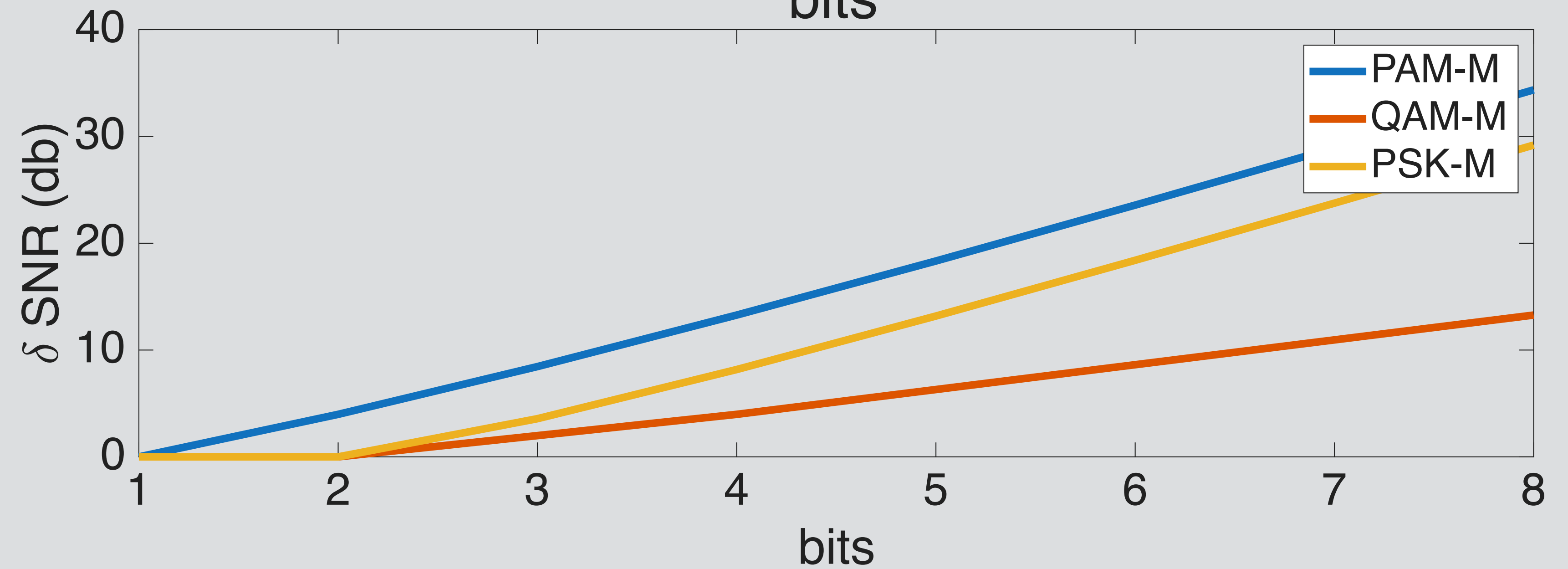
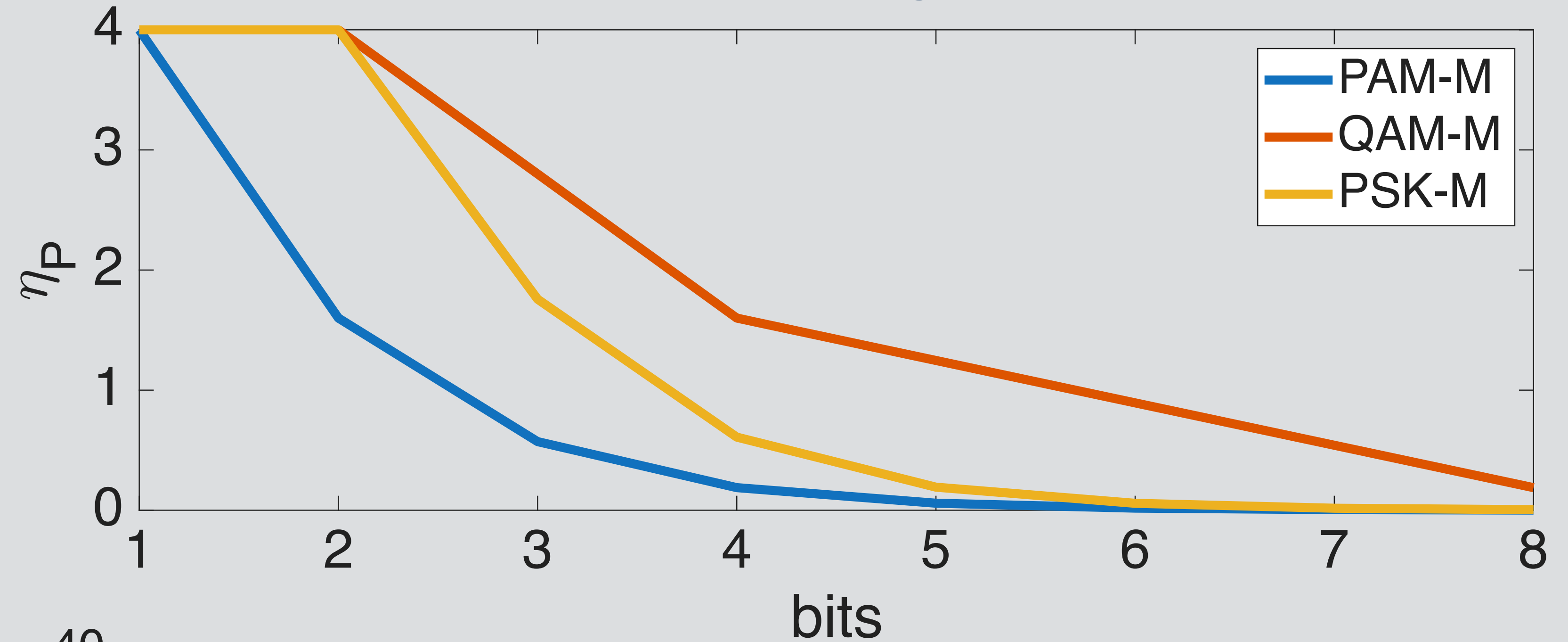
Power Efficiency

$$\eta_{\text{M-PAM}} = 12b/(2^{2b} - 1)$$

$$\eta_{\text{M-QAM}} = 3!b/(2^b - 1)$$

$$\eta_{\text{M-PSK}} = 4b \sin^2(\pi/2^b)$$

$$\delta\text{SNR} = 10 \log_{10} \frac{\eta_{\text{QPSK}}}{\eta_{\text{other}}}$$



Examples

Q. For a passband communication system operating at 40Mbps:
What is the bandwidth if we use QPSK with $\beta = 0.25$

A:

$$\eta_B = \log_2 M = 2 \quad \Rightarrow B_{\min} = R_b / \eta_B = 20 \text{ MHz}$$

$$\Rightarrow B = (1 + \beta) B_{\min} = 20 * 1.25 = 25$$

Examples

Q. For a passband communication system operating at 40Mbps:
What is the bandwidth if we use 16QAM with $\beta = 0.25$

A:

$$\eta_B = \log_2 M = 4 \quad \Rightarrow B_{\min} = R_b / \eta_B = 10 \text{ MHz}$$

$$\Rightarrow B = (1 + \beta) B_{\min} = 12.5 \text{ MHz}$$

Examples

Q. For a passband communication system operating at 40Mbps:
QPSK power is 50mW. Estimate TX power needed for 16QAM
for similar reliability as QPSK

A: Assume same noise N_0

QPSK

$$\eta_{\text{QPSK}} \frac{E_b}{N_0} = 4 \frac{E_{b_{\text{QPSK}}}}{N_0}$$

$$P_{\text{QPSK}} = E_{b_{\text{QPSK}}} R_b = 50$$

16QAM

$$\eta_{\text{QAM}} \frac{E_b}{N_0} = 1.6 \frac{E_{b_{\text{QAM}}}}{N_0} = 1.6 \frac{2.5 E_{b_{\text{QPSK}}}}{N_0}$$

$$P_{\text{QAM}} = E_{b_{\text{QAM}}} R_b = 125$$

Examples

Q. For a passband communication system operating at 40Mbps:

For increased rate to 80Mbps, and same reliability how do the bandwidth and power change for QPSK?

A: Bandwidth scales with rate - previously was 25MHz

$$BW = 50 \text{ MHz}$$

η_P is the same, assume noise power the same, so E_b the same

$$P = E_b 2R_b = 100 \text{ mW}$$